What-If / Checklist Analysis – Quick Guide

Offline What-If / Checklist study tool · part of the Process Safety Analysis Toolkit

1 · What it does and where it fits

A **What-If / Checklist analysis** asks “What if …?” questions about an area, a system or a task, and uses a checklist to find the questions the team missed. For each question the team writes down the hazard, the consequence, the safeguards, the risk and a recommendation. The method is the combined What-If / Checklist analysis of CCPS, *Guidelines for Hazard Evaluation Procedures*, 3rd ed. (2008), section 5.2.

HAZID → **What-If / Checklist** → **HAZOP** → **FMECA** → **Fault tree** → **Event tree** → **Bow tie**

It comes after the HAZID and before the HAZOP. It is quicker and broader than a HAZOP and it suits **tasks and procedures** (tanker unloading, purging, light-off, maintenance isolation) and **good-practice checks**. Many questions are closed here; the ones that need a systematic deviation analysis go on to a HAZOP node.

2 · The five CCPS steps in the tool

Step	Where in the tool
1 · Prepare the review	*Study details & team*, *Process safety info*, *HAZID register* (File ▸ Import HAZID register), *Areas, functions & tasks* – divide the study into physical areas, functions or tasks and choose the checklist packs for each.
2 · List What-If questions	*What-If worksheet* ▸ **Add What-If question**. Brainstorm first: start each question with “What if …”.
3 · Use the checklist to cover gaps	*Checklist*: answer each item **Yes**, **No**, **N/A** or **?**. For a **No** or **?**, click → **What-If** – the question is added to the worksheet with the library prompt.
4 · Evaluate each question	Click a worksheet row: hazard, consequence, safeguards (mark an IPL), severity × likelihood before and after, recommendations, HAZID items examined, and the **next step**.
5 · Document the results	*Reports & export*: print / PDF report, Excel workbook, backup file, and the hand-over to the HAZOP.

3 · The next step for each question

Next step	When to use it
Closed at What-If	The safeguards are adequate, or a recommendation fixes the gap.
Send to HAZOP	The scenario needs a systematic deviation analysis. Write the node (N01) or deviation (N01-06) in *Goes to*.

Send to quantified study	The protection must be quantified – fault tree, event tree or bow tie (e.g. FT-01).
Further study / information	The team needs data or a specialist before it can decide.

> **Hand-over to HAZOP** lists the questions sent to each node. **Export to HAZOP study file** makes a `.hazop.json` for the HAZOP Study Manager with the nodes, deviations, causes, consequences, safeguards (with IPL claims), risk and recommendations already filled in – the HAZOP team then adds the parameter and guideword.

4 · The checklist library

150 checklist items, each with a What-If prompt, in 12 categories (materials, equipment, layout, relief, instruments and trips, utilities, operations, start-up and shutdown, maintenance, human factors, external events, emergency response). Packs: **GEN** generic (every area), **FG** fuel gas networks and furnaces, **LPG** storage and transfer, **NH3** ammonia and fertiliser plants. Add your own items with *Add custom checklist item*. The checklist helps the team – it is not a code-compliance list.

5 · Saving and Excel

- Work is kept automatically in the browser. **File ▸ Save study as…** links a `.whatif.json` file that is updated after every change (Chrome / Edge). Use **Download backup** as a second copy.
- Teams that prefer Excel use `Workbooks/WhatIf_Checklist_Workbook.xlsx` – the same columns, library and risk matrix.

6 · Examples

File ▸ Load example study loads a small hot DRI conveyor study. Complete studies are in the worked examples: the propane tanker-unloading task (`WI-EX-PR-001`), ammonia non-routine tasks and toxic-gas readiness (`WI-EX-AM-001`), and non-routine work on a natural gas network (`WI-EX-GN-001`). Each one links its questions to the HAZID items and to the HAZOP nodes, and each report has a What-If section. Data are illustrative.